

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

6th Semester of B. Pharm. Examination

University Theory Examination May 2017

PH326.3 Production of Bulk Drugs and Drug Intermediates- I

Date: 10.05.2017, Wednesday Time: 10.00 a.m. to 12.00 noon Maximum Marks: 50

Instructions:

1. Section wise separate answer books to be used.
2. Figures to right indicate marks.
3. Draw neat labeled sketches wherever necessary.

SECTION- I

Q-1 Answer the followings:

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|---|----|
| A. What is green Chemistry? | 01 |
| B. What do you mean by process chemistry? | 01 |
| C. Give any two examples of Phase Transfer Catalysts normally used. | 01 |
| D. Define term: Immobilized solvents | 01 |
| E. Enlist documentation and record as per WHO GMP for API. | 01 |

Q-2 Attempt any **TWO** of the following

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|---|----|
| A. Define Synthone and give general rules and guidelines for disconnection with suitable example. | 05 |
| B. Explain good sampling procedure for starting materials. Give significance of reserve samples. | 05 |
| C. Explain the principles of green synthesis. | 05 |

Q-3 Attempt any **TWO** of the following

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|--|----|
| A. Give synthone approach for any two drugs. | 05 |
| B. Explain ideal conditions for reaction designing and set up. | 05 |
| C. Explain in brief ten principles of GMPs. | 05 |

SECTION- II

- Q-4 Answer the followings:
- A. Define solvent extraction. 01
 - B. What do you mean by reflux ratio? 01
 - C. Which are the necessary conditions for good separation of liquids by fractional distillation? 01
 - D. Define evaporation. 01
 - E. Give examples of filter aids and explain their mechanism. 01
- Q-5 Attempt any **TWO** of the following
- A. Explain continuous extraction of solids with a diagram of suitable apparatus. 05
 - B. Add a detailed note on type of distillation stills for rectification. 05
 - C. Add a note on induced crystallization. 05
- Q-6 Attempt any **TWO** of the following
- A. Draw schematics for batch fermentation process and explain in brief steps involved in batch fermentation. 05
 - B. Explain with a process flow diagram the batch manufacturing process of drug substances with a suitable example. 05
 - C. Add a note on continuous liquid- liquid extraction. 05